

SQL Interview Q&A; — Project Deep Dive

Day 7 of 7 — Explaining Your SQL Reasoning

How to Use This Guide

These questions probe how you think about SQL, not just syntax recall — the kind of follow-up an interviewer asks after you've already written a working query. Pair this with [Day-3_SQL/Interview_SQL_Questions.pdf](#), which covers core SQL syntax and concepts.

1. How did you decide which SQL query to write for each business question?

I started from the business question, not the schema — e.g., 'top customers by lifetime spend' immediately implies GROUP BY Customer_ID with SUM(Total_Amount) and ORDER BY ... DESC, so I work backward from the answer shape I need.

2. Why use a CTE for the repeat-vs-one-time customer analysis instead of a subquery?

A CTE named the intermediate result (customer_orders) so the final SELECT reads almost like English: pick the customer type, sum their revenue, compute their share. Nesting the same logic as a subquery would work identically but be harder for a reviewer to follow.

3. How would you scale these queries if the table had 500 million rows instead of 3,000?

Add indexes on the columns used in GROUP BY/WHERE most often (Order_Date, City, Brand, Customer_ID), partition the table by month given the date-range queries, and consider pre-aggregating monthly rollups into a summary table rather than scanning raw transactions every time a dashboard refreshes.

4. What's the risk of using SELECT * in a project like this?

It pulls unnecessary columns, obscures exactly what a query depends on, and breaks silently if the schema changes — I named every needed column explicitly in analysis_queries.sql so each query documents its own dependencies.

5. How did you compute revenue lost to discounting per month?

I compared gross revenue before discount (Price x Quantity) against actual Total_Amount for the same rows, summed both by month, and took the difference — that isolates the discount's dollar cost from the item price itself.

6. Why GROUP BY brand_tier instead of GROUP BY Brand for the return-rate analysis?

The business question was about premium vs. budget, not individual brands, so grouping by a derived tier (via a CASE expression) answers the actual question in one summarized view instead of 12 separate

rows the stakeholder would have to average manually.

7. How would you test that your SQL results match your Python/Excel results?

I ran the same aggregation (e.g., total revenue, orders by city) in all three tools and diffed the numbers — any mismatch usually means a filter, date boundary, or duplicate-counting issue in one of them, and catching that before presenting saves credibility.